MapModes Final Project Report

Project Title: Mobility Analytics for Walkability and Bikeability in Cleveland

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Abstract:
Active mobility: walking, biking, and micromobility, has become essential to shaping healthy, sustainable cities. Despite Cleveland's efforts to expand bike lanes, mobility hubs, and pedestrian-friendly infrastructure, the city's walkability and bikeability remain relatively low. Our project develops a data-driven scoring framework that evaluates pedestrian and cycling accessibility across Cleveland at both tract-level and route-level resolution. Combining datasets from the U.S. Census, OpenStreetMap, GBFS bike-share feeds, Cleveland Moves Vision Plan, and digital elevation models, we compute three metrics: Bike Score, Walk Score, and Route-Level Mobility Score. Our methodology integrates geospatial analytics, inverse distance decay functions, OSMnx routing, zonal elevation statistics, and weighted amenity scoring. The outputs are visualized through choropleths, interactive maps, and route evaluations, providing a comprehensive view of where Cleveland's active mobility infrastructure succeeds and where major gaps remain.

Live Demo: http://mapsmodes-dev-frontend.s3-website.us-east-2.amazonaws.com/

Introduction and Background

The United States is heavily reliant on cars for commuting to and from work. It is also known as a country that lacks extensive public transportation in many urban areas. We wanted to explore how the facilities for alternative methods of transportation are faring in the United States, starting in the city of Cleveland, Ohio. This desire was motivated by increasing public and policymaker sentiment towards making cities more bikeable, walkable, and possessing effective public transportation methods. Cleveland reflects this trend: despite the presence of Bird and Veo micromobility systems and several bikeway corridors, the percentage of active bicyclists remains low. PeopleForBikes ranks Cleveland 528th in the U.S. and 841st overall, citing poor network connectivity and insufficient access to essential destinations.

Cleveland Moves, a recent citywide initiative, proposes to expand bike lanes, improve crosswalks, update signal timing, and install mobility hubs. Our project evaluates how well these improvements could enhance mobility by creating unified indices for walkability and bikeability.

There are multiple datasets that provide information on alternative methods of transportation in the United States, yet there is no single way to view this information and to visualize it. Our project uses multiple datasets to visualize how walkable, bikeable, and accessible public transportation is in Cleveland.

We focus on three core questions:

1. How bikeable is each region of Cleveland based on infrastructure, geography, and amenity access?
2. How walkable is each street corner based on proximity to important points of interest (POIs)?
3. How friendly is a chosen route for walking or biking compared to alternative paths?

Means of Transportation

We used the B08301 - Means of Transportation dataset from the U.S. Census Bureau's American Community Survey to compute the proportion of workers using different transportation modes for commuting.

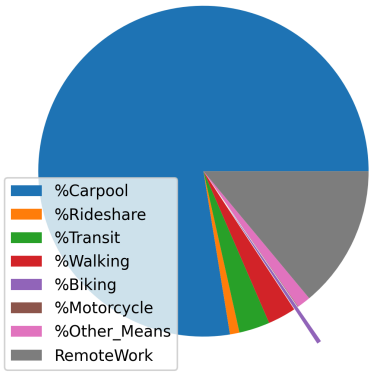


Fig. 3. Distribution of Transport in Cuyahoga County, OH

Despite the presence of several bikeshare and scooter-share companies, such as Bird and Veo, in Cleveland, the percentage of active bikers remains low compared to other urban cities. Based on PeopleForBikes' analysis of Cleveland's bike network, it ranked 24th in Ohio, 528th in the U.S., and 841st out of 2901 cities. Cleveland's bike network is insufficient because it lacks access to residents' homes and neighborhoods, jobs and schools, public transit, and core services such as hospitals and grocery stores.

Bike Score

To improve the bike network and encourage higher levels of bike commuting, the City of Cleveland is launching Cleveland Moves, a mobility plan to expand the bike, walk, and transit network over the next three years. Their goals are to build 50 miles of high-comfort bikeways, modify signal timing, improve crosswalks, convert parking meter poles into bike racks, and install more mobility hubs.

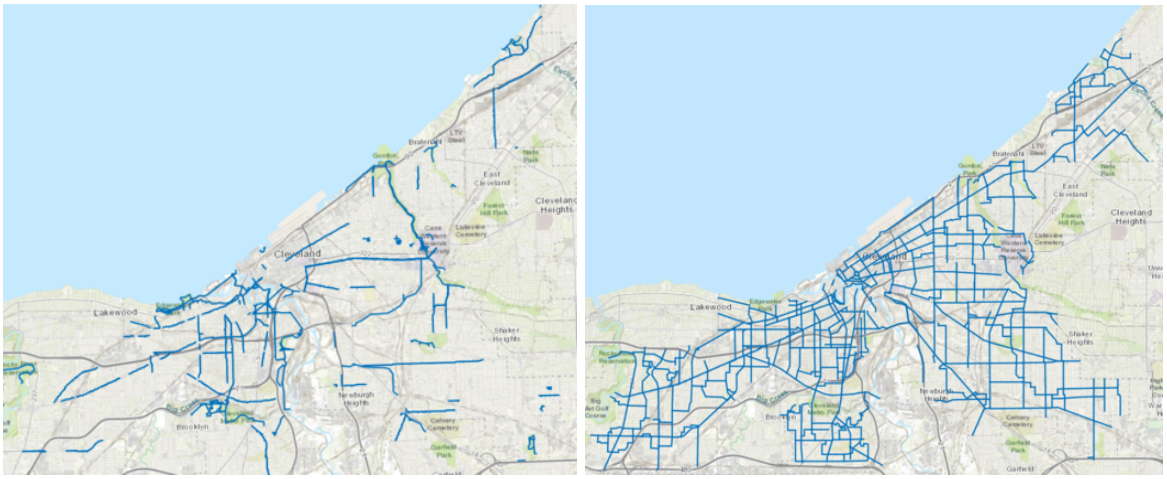


Fig. 4. Existing Bike Network versus Cleveland Moves Vision Bike Network

We developed a bike score metric to evaluate the area's bikeability based on seven factors: bike lane density, intersection density, percentage of cyclists, slope, proximity to amenities, rental bikes, and mobility hubs. Bike lane density is the total length of bike lanes divided by the total street length within a specific region

$$bike\ lane\ density\ =\ \frac{total\ bike\ lane\ length}{total\ area}\ .$$

It is important to assess the bikeability because a dense bike network makes it easier and safer for cyclists to reach their destinations. With more bike lanes, there will be more access and route options for amenities. To compute the total bike lane length per tract, we created a new dataframe that merges the Cleveland Moves Vision Bike Network dataset with the Cleveland shapefile, using the GEOID identifier as a key.

Intersection density is the number of intersections per unit area, which measures the connectivity of a street's network. Higher intersection density indicates better connectivity, enabling shorter distances and more direct routes. The intersection density was computed by dividing the number of intersections by the tract's area. Using the OpenStreetMap package to obtain a dataset of Cleveland's streets and intersections, we created and organized a new dataframe to merge the intersections dataset and the shapefile to determine the number of intersections per tract, using the GEOID as a key. Using this new dataframe, we calculate the intersection density.

The next metric that we calculated was slope, which measures how flat or steep the route is. The smaller the slope of the route, the easier route is to travel. Using Cleveland's Digital Elevation Model (DEM) file and the rasterio package, we obtained the elevation statistics and

used the *zonal_stats* function to calculate the min and max elevation values per tract. We then used the following formula to calculate the slope metric, the rate of change in elevation,

$$slope = \frac{elevation_{max} - elevationn_{min}}{lane\ length}.$$

The next metric we obtained was the percentage of cyclists per tract from the B08301 (Means of Transportation) dataset. It's an important metric because, as the number of cyclists increases, more people are encouraged to follow their example. Most people judged an activity's safety and accessibility based on its popularity.

Proximity to amenities, rental bikes, and mobility hubs are the last factors for the bike score. They are crucial because they determine whether to choose to drive, walk, or bike. Proximity to amenities increases the likelihood that people will choose active modes of transportation, such as biking and walking. The OpenStreetMap package also provides a list of amenities for a given address or location. The amenities dataset was then organized into the following categories: education, grocery, food, general stores, money, public service, healthcare, health, entertainment, retail, and transportation. The categories were then re-weighted based on necessity:

5 - Education, Grocery, Healthcare, Transportation

4 - General stores, Public Service

3 - Retail, Food, Money

2 - Entertainment, Health

1 - Other.

To calculate the amenity score for a given tract, the amenity's weights were summed and then divided by the tract's total number of amenities.

The General Bikeshare Feed Specification (GBFS), an open data standard for shared mobility of bikes, scooters, and mopeds, was used to obtain the list of available Bird bikes for rent. The City of Cleveland Office of Urban Analytics and Innovation (Urban AI) also provides a dataset of mobility hubs for the Cleveland and Cuyahoga County. With these datasets, the rental bike and mobility hubs scores were calculated by dividing the number of bikes and hubs in a given tract by the tract's area.

Each factors were then normalized, assigned weights to represent its importance, and combined to calculate the bike score *B* for each tract

$$B = 0.25x_{bike\ lane} + 0.18x_{interst.} + 0.18x_{\%\ biking} + 0.18x_{amenity} + 0.08x_{hub} + 0.08x_{bike} + 0.08x_{slope}.$$

Fig. 5. Bike Scores based on the Cleveland Moves Vision Bike Network

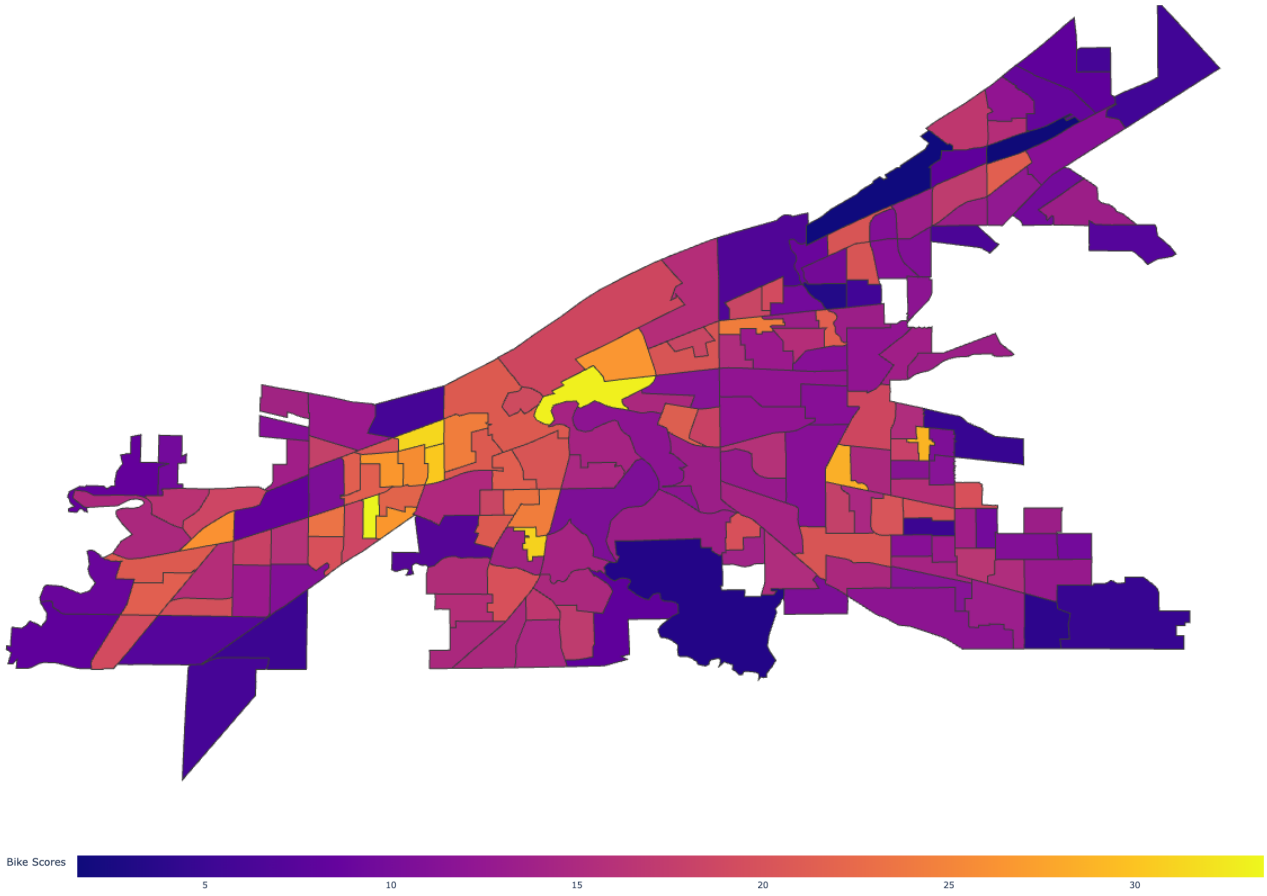


Fig. 6. Choropleth of Bike Scores

After computing regional bike scores, we can extend the methodology to create a route-level scoring algorithm that evaluates an address and bike route based on distance and travel time. The primary difference was that an inverse distance decay function was implemented for the proximity to amenities, mobility hubs, and available rental bikes factors to assign a stronger influence to factors closer to the point of interest and decrease influence as distance increases

$$w = e^{-0.001*distance}.$$

Walk Scores

Our walk scores remained consistent for the duration of this project. We built a few different functions to calculate these scores: one that could take any address and calculate its score, and another that could take any set of coordinates and return the corresponding score. The

coordinate-based function presented a surprising number of issues, considering that the first step of the address-based function was to convert the address into coordinates.

Once we got the coordinate function working, we used it to calculate the walk score displayed on the map. To do that, we created a list of every street corner in Cleveland and calculated a score for each. To gather all the data, we used OpenStreetMaps (OSM) and the corresponding Python package, OSMnx. OSM is an open-source alternative to the Google Maps API that allows the public to label every feature on the map, from apartment buildings and houses to bus stops and streets, to natural features like viewpoints or lakes. From this source, we were able to get every street corner in Cleveland, as well as a list of every "point of interest" within a certain distance of each corner.

We then calculated the walk score for each street corner by counting the number of destinations reachable within a given time. The score was weighted by both the travel time and the "utility" of each point of interest. As we did not have data to objectively assess the utility of each point, we estimated its usefulness using the following table.

place	5_min	10_min	20_min	place	5_min	10_min	20_min	place	5_min	10_min	20_min
bar	5	5	3	supermarket	10	8	2	playground	1	1	1
biergarten	5	5	3	religious	1	1	1	swimming_pool	1	1	1
cafe	8	5	3	cathedral	1	1	1	track	1	1	1
fast_food	5	5	3	chapel	1	1	1	water_park	1	1	1
food_court	3	3	1	church	1	1	1	alcohol	1	1	0
ice_cream	5	3	1	mosque	1	1	1	beverages	1	1	0
pub	5	5	3	shrine	1	1	1	butcher	1	1	0
restaurant	8	6	4	synagogue	1	1	1	cheese	0.5	0.5	0
college	1	1	1	temple	1	1	1	chocolate	0.5	0.5	0
kindergarten	3	3	0	museum	2	2	2	coffee	2	1	0
language_schoo	1	1	1	train_station	7	4	1	confectionery	1	1	0
library	4	4	1	sports_hall	1	1	1	convenience	6	4	1
dancing_school	1	1	1	sports_centre	1	1	1	deli	5	3	1
research_institut	1	1	1	stadium	1	1	1	food	6	6	3
music_school	1	1	1	castle	0.1	0.1	0.1	greengrocer	5	2	1
school	8	5	2	bakery	6	4	2	health_food	1	1	1
university	1	1	1	brewery	1	1	1	nuts	1	1	0
ferry_terminal	1	1	0	confectionery	1	1	1	pasta	1	1	0
bus_station	1	0	0	dressmaker	0.5	0.5	0.5	pastry	1	1	1
atm	1	0	0	embroiderer	1	1	1	seafood	1	1	0
bank	5	3	1	electronics_repa	1	1	1	department_stor	8	4	1
clinic	2	2	2	handicraft	0.5	0.5	0.5	general	6	4	1
dentist	2	2	2	jeweller	1	1	1	mall	8	4	1
doctors	2	2	2	photographer	1	1	1	supermarket	10	8	2
hospital	2	2	2	pottery	1	1	1	clothes	4	3	1
pharmacy	3	3	2	tailor	1	1	1	fashion_accesso	1	1	1
veterinary	0.5	0.5	0.5	audiologist	1	1	1	jewelry	1	1	1
arts_centre	1	1	1	counselling	1	1	1	beauty	1	1	1
casino	0.5	0.5	0.5	nurse	1	1	1	cosmetics	1	1	1
cinema	1	1	1	optometrist	1	1	1	hairdresser	1	4	1
community_cent	0.5	0.5	0.5	physiotherapist	1	1	1	massage	1	1	1
conference_cent	0.5	0.5	0.5	podiatrist	1	1	1	optician	1	1	1
events_venue	0.5	0.5	0.5	amusement_arc	1	1	1	electronics	1	1	1
music_venue	1	1	1	dance	1	1	1	toys	1	1	1
theatre	1	1	1	dog_park	1	1	1	attraction	1	1	1
post_office	1	1	1	fitness_centre	1	1	1	gallery	1	1	1
internet_cafe	1	1	1	ice_rink	1	1	1	museum	1	1	1
marketplace	2	2	2	miniature_golf	1	1	1	theme_park	1	1	1
place_of_worshi	2	2	2	park	1	1	1	zoo	1	1	1
commercial	2	2	2	pitch	1	1	1	viewpoint	1	1	1

Fig. 7. POIs common in the OSM dataset

The results from this process ranged from as low as 3 in a suburban residential area to over 2000 in downtown Cleveland. Because those are relatively meaningless numbers to someone who didn't build them or research them, we had to normalize the results. We normalized the results to get to a 0-100 scale by using the equation:

$$\min(100, \text{walk_score}/100)$$

which works for the roughly 70,000 points we have within Cleveland, which is the scope of our project at this moment.

We used these normalized walk scores to make a map of the city that could be zoomed in or out to look at the walk score of a certain region. For example, the university circle region looks like this:

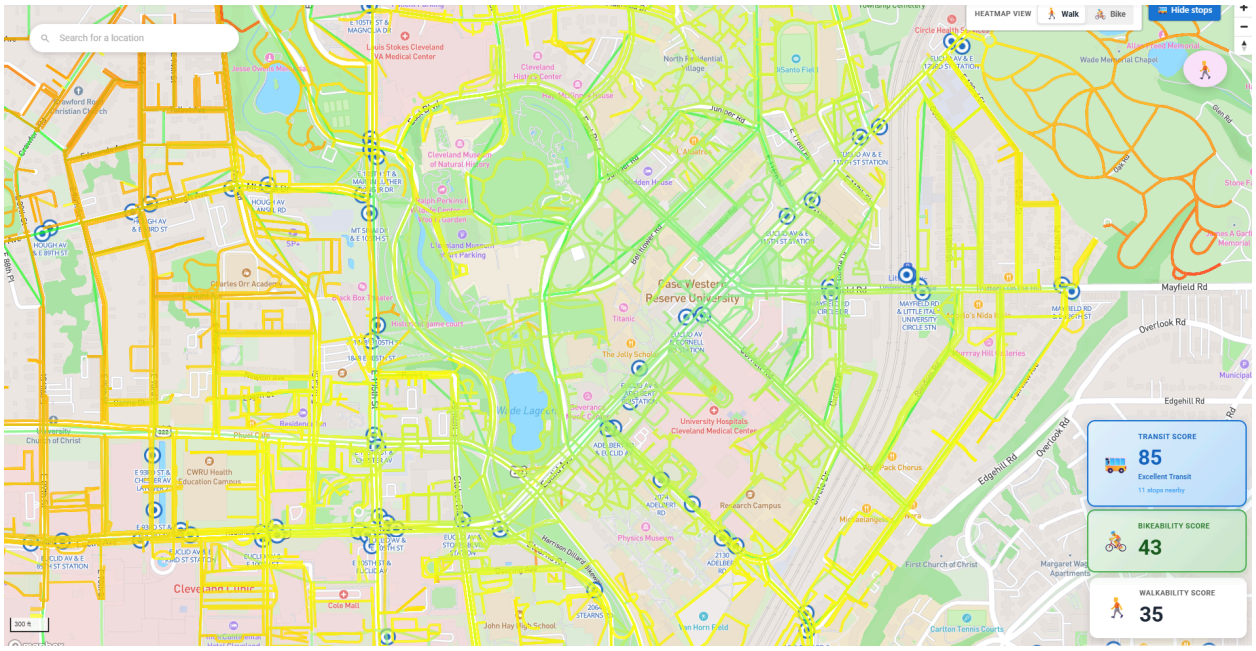
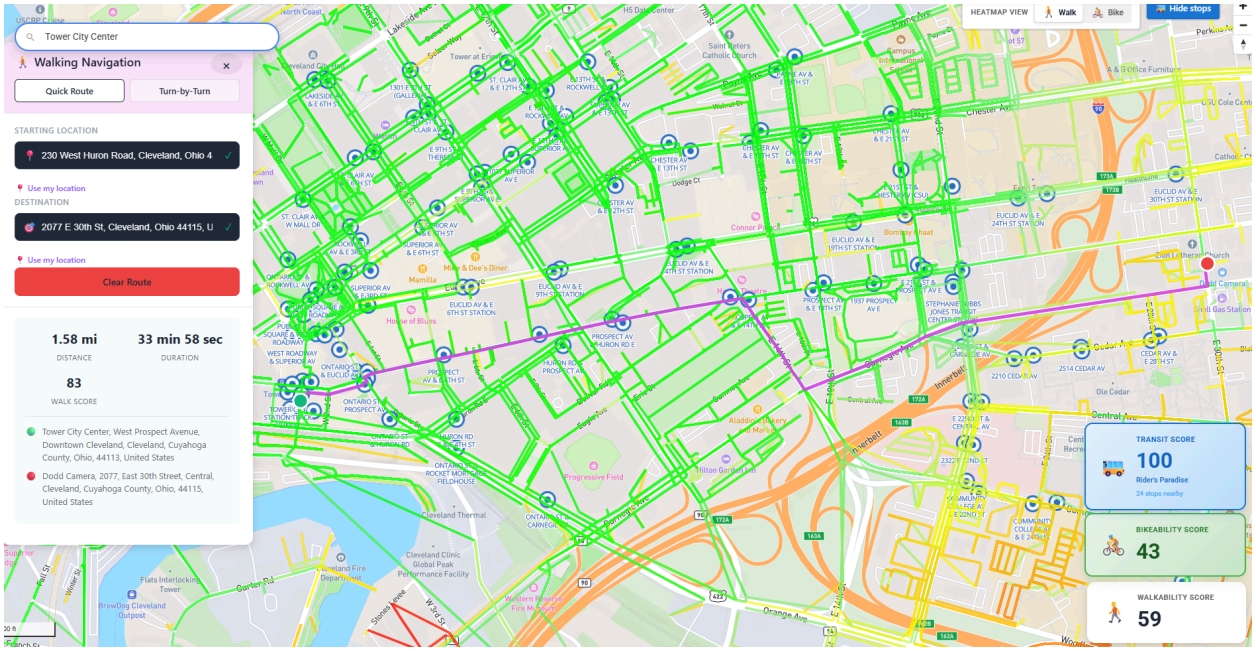


Fig. 8. University Circle Walkability Map

Which makes sense that the "hub" around Euclid and up Mayfield through Little Italy appear as very walkable regions, but the cemetery, and surrounding areas heading towards town are far less walkable. This also shows the limitation that we only have data from Cleveland, because the top of the hill, in Cleveland Heights, doesn't appear on our map.

We also built a feature that allows users to get walking directions from any two points within the city. It also calculates how easy and walker friendly the path along the way is. For

example, going from Dodd Camera to Tower City gives this path:



Much more walker friendly in downtown than in the surrounding areas, as shown by the walk score of 83 along the route compared to 59 on the screen as a whole. This is almost exactly what we were expecting and looking to do when we set out to make this project.

Transit Score

To complement the walkability and bikeability indices, we introduce a Transit Score that quantifies how well an area is served by public transportation infrastructure. The score ranges from 0 to 100 and is composed of three components: proximity to transit stops, diversity of reachable routes, and the accessibility of the nearest stop. This formulation emphasizes both physical closeness to transit and the network richness that affects travel utility.

$$Transit\ Score = S_p + S_r + S_c$$

where:

- S_p: Stop Proximity Score (0-40 points)
- S_r: Route Coverage Score (0-30 points)
- S_c: Closest Stop Score (0-30 points)

Each component captures a distinct aspect of transit accessibility: density of stops, diversity of service, and ease of reaching the closest stop. Together, they approximate a user's effective access to Cleveland's RTA system.

Stop Proximity Score

The Stop Proximity Score evaluates how many transit stops fall within a typical pedestrian-accessible radius: set to 500 meters, consistent with walkability literature. Let denote n_{stops} the number of stops in this radius. The score increases linearly until it saturates at 40 points:

$$S_p = min(40, 4 \times n_{stops})$$

This reflects the intuition that areas with many nearby stops are more flexible and resilient to schedule variations, although the marginal benefit of additional stops eventually plateaus.

Route Coverage Score S_r

Stop density alone does not guarantee useful service; riders also care about the diversity of bus routes available. Let n_{routes} denote the number of distinct RTA bus or rail routes accessible within the same 500 m radius. We compute:

$$S_r = min(30, 3 \times \lfloor n_{routes}/2 \rfloor)$$

This heuristic awards more points to locations where users can board multiple lines, enabling faster transfers, greater geographic reach, and more robust travel options. The score saturates after approximately twenty accessible routes, reflecting diminishing utility beyond that point.

Closest Stop Score S_c

The final component quantifies the ease of reaching the nearest stop. Extremely close stops (≤ 100 m) award full points, while accessibility declines linearly until 500 m, beyond which service is effectively considered inaccessible.

$$\begin{cases} 30 & d_{\text{closest}} \leq 100 \text{ m}, \\ 30 \times \left(1 - \dfrac{d_{\text{closest}} - 100}{400}\right) & 100 < d_{\text{closest}} \leq 500 \text{ m}, \\ 0 & d_{\text{closest}} > 500 \text{ m}. \end{cases}$$

The resulting Transit Score captures Cleveland's transit accessibility across three dimensions: stop availability, network richness, and physical ease of access. Together, these components allow us to classify areas into high-access transit zones, moderate-access areas, and transit deserts, regions where the nearest stop lies beyond 800 m and where service diversity is low.

Design Specifications

The design of MapsModes is driven by three goals: (1) provide quantitative measures of walkability, bikeability, and transit access at neighborhood scale, (2) expose these metrics through a reproducible cloud pipeline, and (3) make the results explorable through a simple web interface.

Functional Specifications

- Mobility Scoring
    - Compute walk scores for intersections and grid cells based on intersection density and proximity to weighted points of interest.
    - Compute bike scores for census tracts using bike lane density, intersection density, elevation/slope, cycling mode share, amenity access, mobility hubs, and shared bikes.
    - Compute transit access scores by measuring distance to the nearest bus stop and classifying areas into excellent access, moderate access, and transit deserts (> 800 m).
- Data Integration
    - Ingest and clean OpenStreetMap (OSM) data for streets, intersections, and amenities.
    - Ingest GTFS data for RTA routes and stops.
    - Ingest ACS census data for demographic context.
    - Store processed datasets and derived scores in a consistent geospatial schema.
- Interactive Web Application
    - Serve tiled map layers (walk, bike, and transit coverage) to a React + Mapbox frontend.
    - Support basic interaction: panning, zooming, and hovering/clicking to reveal scores for specific locations.

    - Provide separate views for transit coverage, walkability, and bikeability.

System Architecture Specifications

- Backend and ETL
    - Use scheduled AWS Lambda ETL jobs to fetch, clean, and transform OSM, GTFS, and ACS data.
    - Store raw and intermediate datasets in S3.
    - Use Python (GeoPandas, NetworkX, rasterio) to compute network graphs, distances, and mobility scores.
    - Publish curated GeoJSON/vector tiles and summary tables for fast access.
- API Layer
    - Expose read-only endpoints (e.g., */scores*, */tiles*, */search*) via a FastAPI application running on Lambda behind API Gateway.
    - Ensure responses are suitable for low-latency use in a browser map client.
- Frontend
    - Implement a single-page React application using Mapbox GL JS for visualization.
    - Load background basemap tiles from Mapbox and overlay mobility layers from the project's API/S3 tiles.
    - Provide simple controls to toggle layers and view statistics for selected regions.

Non-Functional Specifications

- Performance
    - ETL jobs must complete within a single Lambda execution window per dataset ingested.
    - Map tiles and scores should load in under ~2 seconds on a typical campus network connection.
- Scalability
    - Architecture must support re-running the ETL pipeline on updated data without manual intervention.
    - Design should be transferable to other cities by changing configuration and data sources.
- Usability

- Poster and web UI should be interpretable by non-technical stakeholders (planners, students, residents).
- Color scales and legends must clearly indicate relative access (e.g., excellent vs. moderate vs. transit desert).

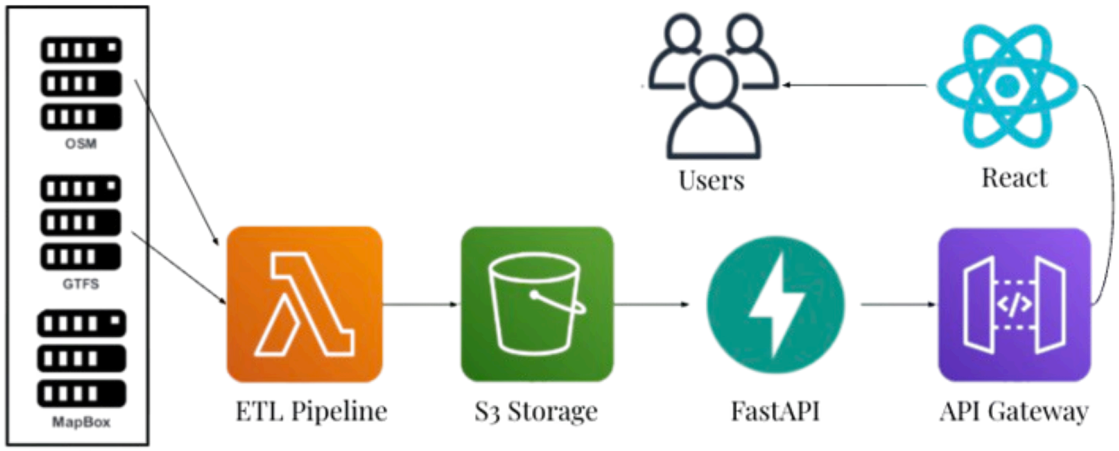


Fig. 10. AWS Software Architecture

Future Work

While the current version of MapsModes delivers a working end-to-end system for Cleveland, several extensions could significantly increase its utility and realism:

Richer Transit Metrics*:* Move beyond stop distance to incorporate service frequency, headways, and time-of-day coverage using full GTFS schedules. This would allow us to distinguish between areas that merely have nearby stops and those that truly have reliable, frequent service.
Safety and Comfort Layers*:* Integrate crash data, speed limits, and traffic volume to estimate Level of Traffic Stress (LTS) for cyclists and pedestrians. This would refine the bike score from "can you bike here?" to "can the average person bike here comfortably?"
Sidewalk and Infrastructure Quality*:* Incorporate data on sidewalk presence, width, and condition, either from municipal inventories or from computer vision applied to street-level

imagery. This would improve walk score accuracy, especially in lower-income neighborhoods where infrastructure is inconsistent.

***15-Minute City and Amenity Access**:* Extend the analysis to identify "15-minute neighborhoods" where groceries, healthcare, schools, and parks are all reachable within a short walk or transit ride. This would connect our work more directly to equity and health outcomes.

***Multi-City and Regional Expansion**:* Generalize the pipeline so it can be configured for other cities or for all of Cuyahoga County. Parameterizing bounding boxes, transit feeds, and data sources would make MapsModes a reusable framework rather than a Cleveland-only tool.

***User-Facing Features**:* Add features such as saving favorite locations, exporting reports for specific neighborhoods, or allowing planners to simulate the effect of adding new bus stops or bike lanes and immediately seeing how scores change.

***Machine Learning and Prediction**:* Use clustering or predictive models to classify neighborhood mobility "types" and forecast how proposed infrastructure projects might change walk, bike, and transit accessibility over time.

Conclusion

MapsModes was developed to answer a simple but important question: how walkable, bikeable, and transit-accessible is Cleveland, really? By combining OpenStreetMap, GTFS, and Census data in a cloud-based pipeline and exposing the results through an interactive web map, we produced a detailed, quantitative view of mobility across the city.

Our analysis shows strong accessibility in downtown and core corridors such as University Circle, where walk, bike, and transit infrastructure are dense and well-connected. At the same time, we identify large pockets of the city that qualify as transit deserts, located more than 800 meters from the nearest bus stop, and neighborhoods where walk and bike networks are fragmented despite relatively high population density. These findings confirm that Cleveland's reliance on cars is not just cultural but structurally encoded in its infrastructure.

Beyond individual maps and statistics, the main contribution of this project is the end-to-end system: a reproducible ETL pipeline, scoring framework, and web interface that together form a

reusable toolkit for mobility assessment. City planners could use MapsModes to prioritize new routes or bike lanes, community organizations could use it to advocate for better service in underserved areas, and researchers could extend it to related questions such as environmental justice or public health.

Overall, MapsModes demonstrates that relatively accessible open data and modern cloud tools are sufficient to build actionable, city-scale mobility analytics. With future extensions, such as richer transit metrics, safety layers, and 15-minute city analysis, this framework has the potential to become a powerful decision-support tool for making Cleveland and other cities more equitable and less car-dependent.

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